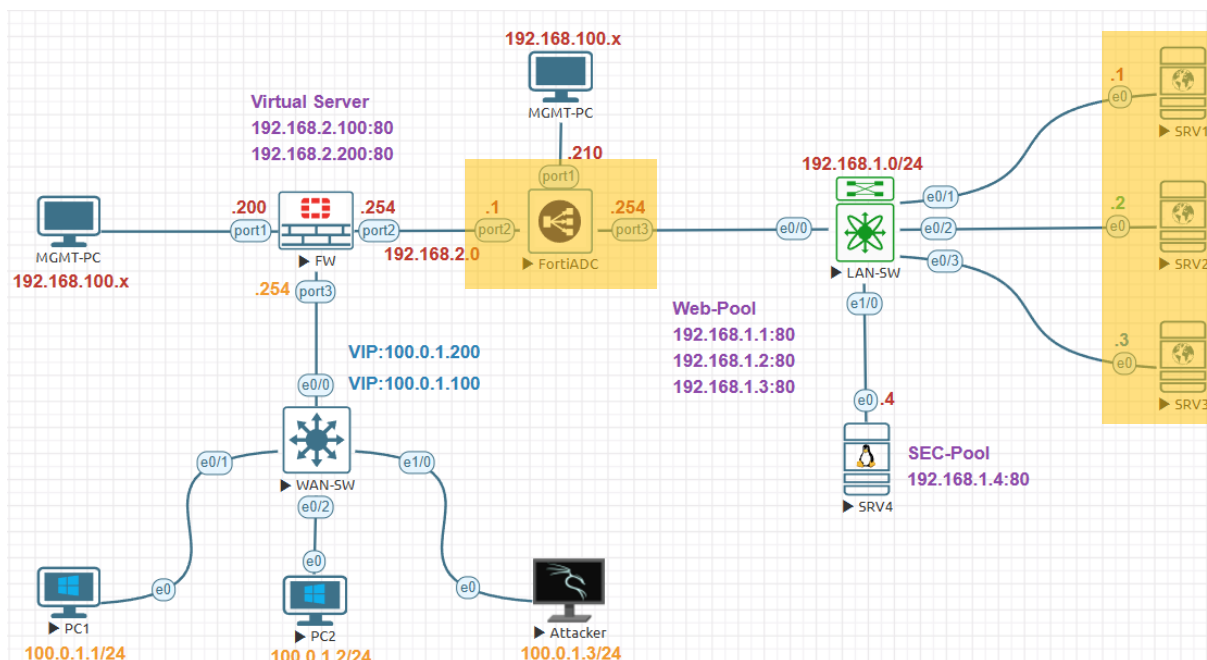


ICMP Health Check Lab:



Management Subnet	192.168.100.0/24
FortiGate Management IP	192.168.100.200
FortiADC Management IP	192.168.100.210
Internal Servers Subnet	192.168.1.0/24
FortiADC External Subnet	192.168.2.0/24
FortiGate Firewall External	100.0.1.0/24
FortiADC Internal IP Address	192.168.1.254
FortiADC External IP Address	192.168.2.1
FortiGate Firewall Internal IP	192.168.2.254
FortiGate Firewall External IP	100.0.1.254
SRV1 IP Address	192.168.1.1
SRV2 IP Address	192.168.1.2
SRV3 IP Address	192.168.1.3
SRV3 IP Address	192.168.1.4
Virtual Sever-1	192.168.2.100
Virtual Sever-2	192.168.2.200
FortiGate VIP-1	100.0.1.100
FortiGate VIP-2	100.0.1.200
External PC1 IP Address	100.0.1.1
External PC2 IP Address	100.0.1.2
External PC3 IP Address	100.0.1.3

Let's create our own custom Health Check Navigate to Shared **Resources>Health Check>Health Check** click on **Create New**

FortiADC FortiADC

Dashboard > Health Check Health Check Monitor Health Check Script

Security Fabric > Delete + Create New + Add Filter

FortiView >

System >

Shared Resources > Health Check

Name	Type
LB_HLTHCK_ICMP	ICMP
LB_HLTHCK_HTTP	HTTP
LB_HLTHCK_HTTPS	HTTPS
LB_HLTHCK_TCP_ECHO	TCP Echo

Showing 1 to 4 of 4 entries 0 rows selected Show 25 entries

Type **Name** in this case **ICMP-HealthCheck** and choose Type **ICMP** click **Save**.

Health Check

Name ICMP-HealthCheck

Type ICMP

General

Destination Address Type IPv4 IPv6

Destination Address 0.0.0.0
Example: 192.0.2.1

Up Retry 1
Default: 1 Range: 1-10 retries

Down Retry 3
Default: 3 Range: 1-10 retries

Interval 5
Default: 5 Range: 1-3600 seconds

Timeout 3
Default: 3 Range: 1-3600 seconds

Save

Next navigate to **Server Load Balance > Real Server Pool > Real Server Pool** tab double click on WEB-Pool to open the setting.

Dashboard > Security Fabric > FortiView > System > Shared Resources > Network > **Server Load Balance** > Virtual Server > Application Resources > Application Optimization > **Real Server Pool**

Real Server Pool Real Server Server SSL

Delete + Create New + Add Filter

Name	Address Type	Health Check
WEB-Pool	IPv4	Enable
SEC-Pool	IPv4	Disable

Showing 1 to 2 of 2 entries 0 rows selected Show 25 entries

Toggle to enable the **Health Check** on right side Double-click to choose **ICMP-HealthCheck** previously created click **Save** to apply the Health Check it will inherited automatically.

Dashboard > Security Fabric > FortiView > System > Shared Resources > Network > **Server Load Balance** > Virtual Server > Application Resources > Application Optimization > **Real Server Pool** > Scripting > SSL-FP Resources

Real Server Pool

Name: WEB-Pool

Address Type: IPv4 IPv6

Type: Static Dynamic

Health Check: ☒ **Health Check**

Health Check Relationship: AND OR

Health Check List: ICMP-HealthCheck

Available Items: LB_HLTHCK_HTTP, LB_HLTHCK_HTTPS, LB_HLTHCK_TCP_ECHO, LB_HLTHCK_ICMP

Direct Route Mode: ☐

Real Server SSL Profile: NONE

Member

Delete + Create New + Add Filter

ID	Name	Address	Health Check
1	SRV1	192.168.1.1	inherited
2	SRV2	192.168.1.2	inherited
3	SRV3	192.168.1.3	inherited

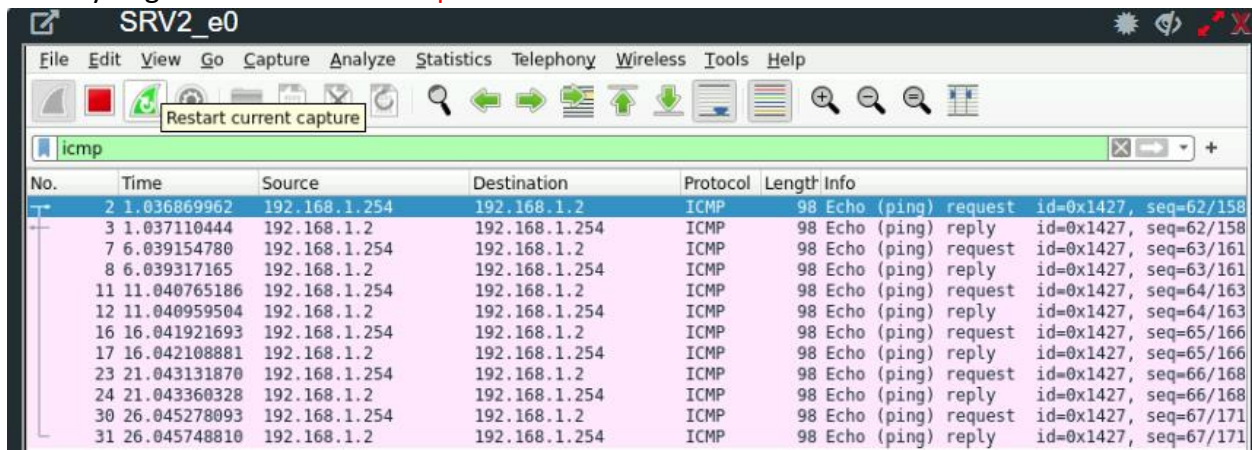
Save Cancel

Verification:

Go to SRV1 login with username & password root/root type **tcpdump icmp**.

```
SRV1 x
root@SRV1:~# tcpdump icmp
tcpdump: verbose output suppressed, use -v or -vv for full protocol
listening on ens3, link-type EN10MB (Ethernet), capture size 262144
17:53:53.388328 IP 192.168.1.254 > 192.168.1.1: ICMP echo request, id=62, seq=62/158
17:53:53.388349 IP 192.168.1.1 > 192.168.1.254: ICMP echo reply, id=62, seq=62/158
17:53:58.390331 IP 192.168.1.254 > 192.168.1.1: ICMP echo request, id=63, seq=63/161
17:53:58.390350 IP 192.168.1.1 > 192.168.1.254: ICMP echo reply, id=63, seq=63/161
17:54:03.391584 IP 192.168.1.254 > 192.168.1.1: ICMP echo request, id=64, seq=64/163
17:54:03.391600 IP 192.168.1.1 > 192.168.1.254: ICMP echo reply, id=64, seq=64/163
^C
```

Alternatively, you can use the built-in Wireshark capture tool in EVE-NG to monitor ICMP traffic directly. Right click one **SRV2 > Capture** choose interface **e0**



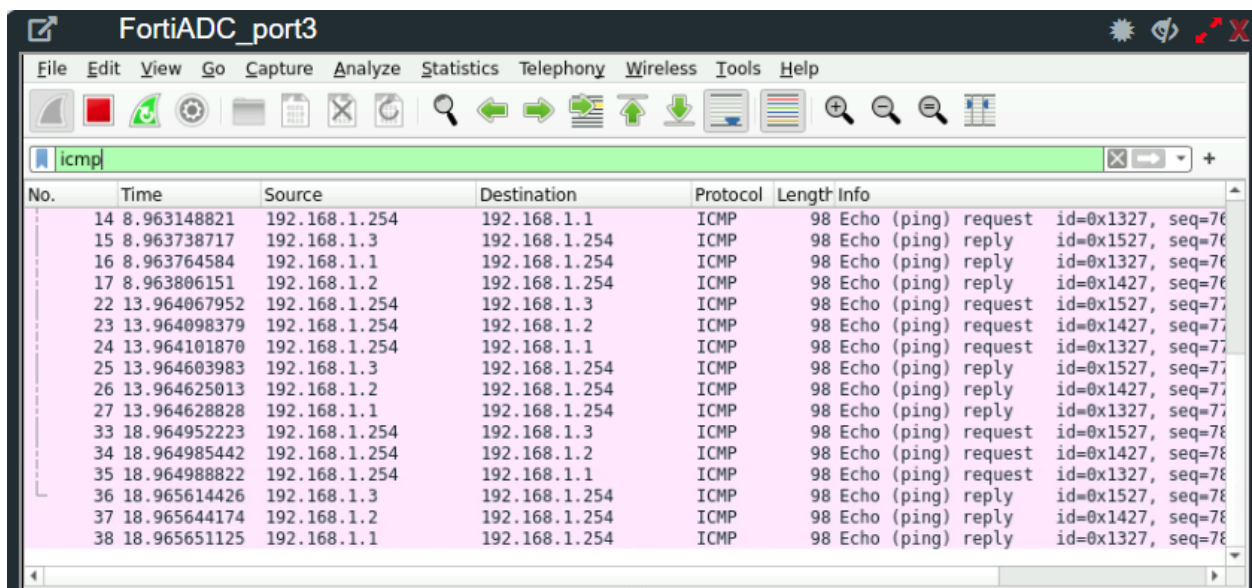
SRV2_e0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Restart current capture

icmp

No.	Time	Source	Destination	Protocol	Length	Info
2	1.036869962	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=62/158
3	1.037110444	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=62/158
7	6.039154780	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=63/161
8	6.039317165	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=63/161
11	11.040765186	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=64/163
12	11.040959504	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=64/163
16	16.041921693	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=65/166
17	16.042108881	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=65/166
23	21.043131870	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=66/168
24	21.043360328	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=66/168
30	26.045278093	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=67/171
31	26.045748810	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=67/171



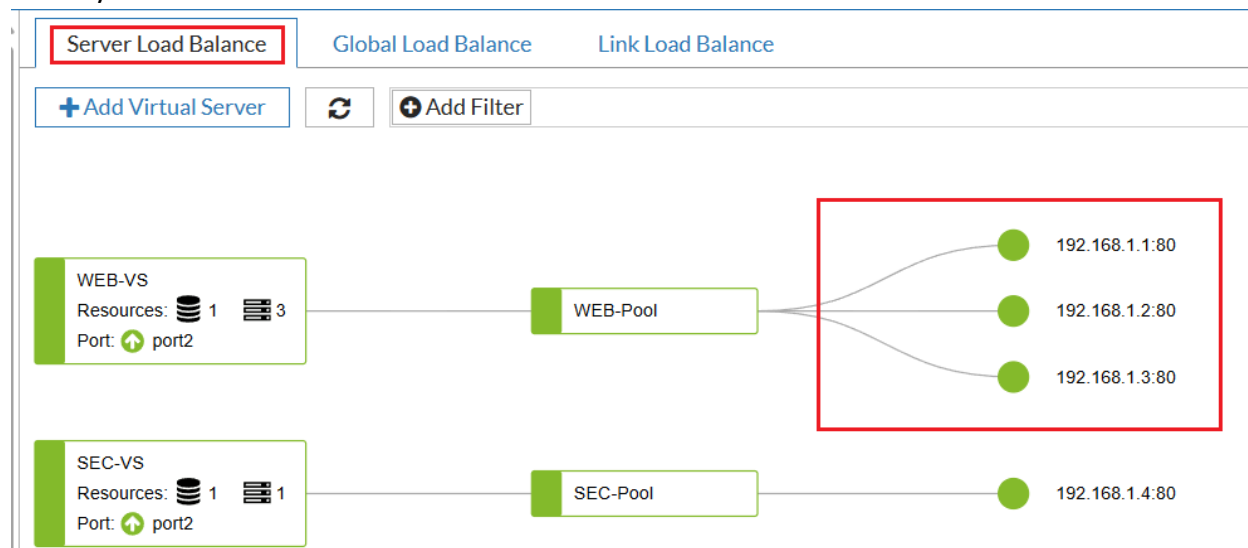
FortiADC_port3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
14	8.963148821	192.168.1.254	192.168.1.1	ICMP	98	Echo (ping) request id=0x1327, seq=76
15	8.963738717	192.168.1.3	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1527, seq=76
16	8.963764584	192.168.1.1	192.168.1.254	ICMP	98	Echo (ping) request id=0x1327, seq=76
17	8.963806151	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=76
22	13.964067952	192.168.1.254	192.168.1.3	ICMP	98	Echo (ping) request id=0x1527, seq=77
23	13.964098379	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=77
24	13.964101870	192.168.1.254	192.168.1.1	ICMP	98	Echo (ping) request id=0x1327, seq=77
25	13.964603983	192.168.1.3	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1527, seq=77
26	13.964625013	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=77
27	13.964628828	192.168.1.1	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1327, seq=77
33	18.964952223	192.168.1.254	192.168.1.3	ICMP	98	Echo (ping) request id=0x1527, seq=78
34	18.964985442	192.168.1.254	192.168.1.2	ICMP	98	Echo (ping) request id=0x1427, seq=78
35	18.964988822	192.168.1.254	192.168.1.1	ICMP	98	Echo (ping) request id=0x1327, seq=78
36	18.965614426	192.168.1.3	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1527, seq=78
37	18.965644174	192.168.1.2	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1427, seq=78
38	18.965651125	192.168.1.1	192.168.1.254	ICMP	98	Echo (ping) reply id=0x1327, seq=78

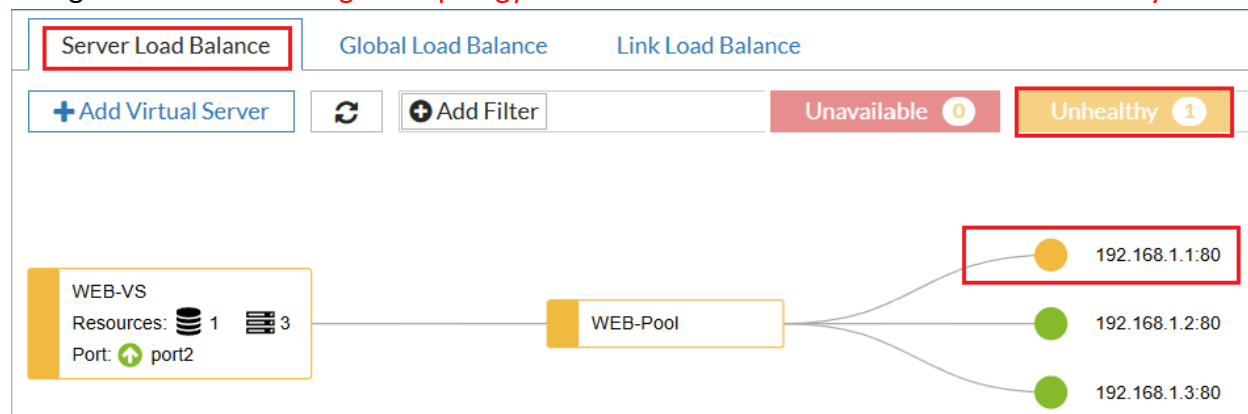
Navigate to **FortiView>Logical Topology>Server Load Balance** all the servers are up green and healthy.



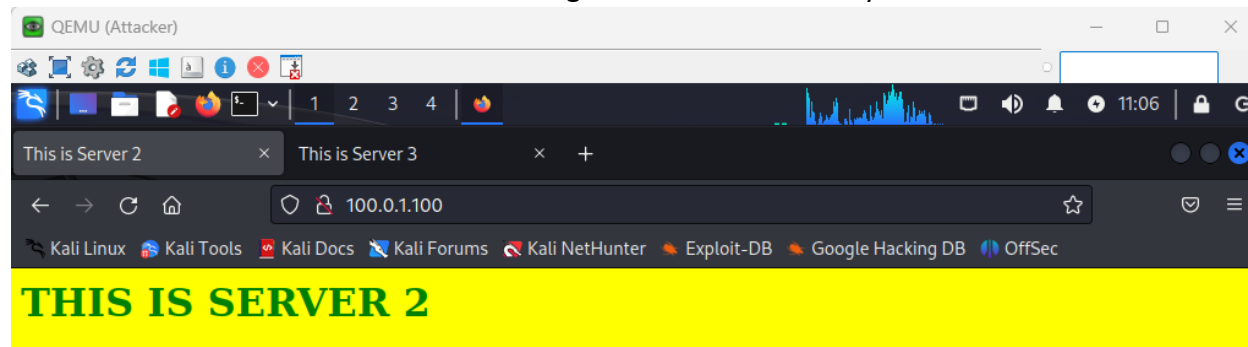
To test the ICMP-Based Health check let's stop one of the Server in this case **SRV1**.

root@SRV1:~# shutdown -h now

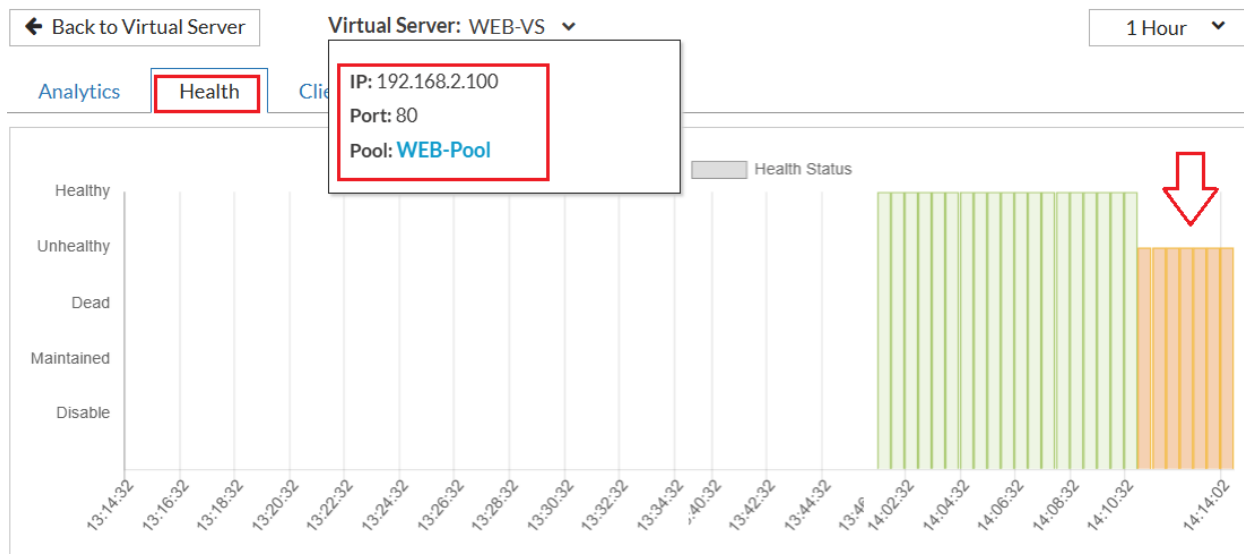
Navigate to **FortiView>Logical Topology>Server Load Balance** this time **SRV1** is unhealthy



Based on the health FortiADC is not sending traffic to **Server 1** only to SERVER 2 and SERVER 3

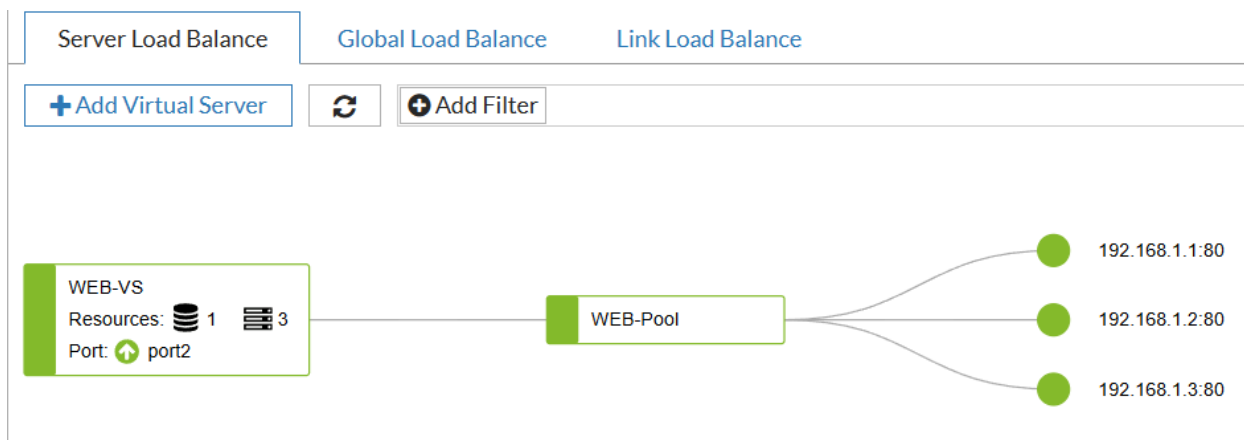


Navigate to **FortiView>Virtual Server** click on the name of Virtual Server in this case **WEB-VS** after that click on **Health** tab to check Health Status the organ color show is unhealthy



Let's start **SERV1**, but this time, stop only the **Web Services** running on it. This approach allows us to simulate a scenario where the server itself is operational and can respond to basic network probes, such as ICMP (ping), but the application or web service is unavailable.

root@SRV1:~# service apache2 stop



In this case ICMP-based health check were used, FortiADC not able to detect the failure on Servers. While the **web service** on servers are down, the server is still capable of responding to ICMP probe packets. Consequently, FortiADC would interpret the server as healthy, as it relies on the ICMP replies for status verification, missing the underlying application-level issue. This highlights the importance of using health checks that are specific to the application's functionality for accurate failure detection.